PSD (Power Spectrum Density)

For WSS RP, Fourier trans of $R(\tau)$

$S(\omega) = \mathcal{F}R(\tau) = \int R(\tau) e^{-j\omega\tau} d\tau$ $\quad S(f) = \int R(\tau) e^{-j2\pi f\tau} d\tau$

$R(\tau) = \mathcal{F}^{-1}S(\omega) = \frac{1}{2\pi}\int S(\omega) e^{j\omega\tau} d\omega$

Avg Power $\bar{P} = E(|X(t)|^2) = R(0) = \frac{1}{2\pi}\int S(\omega) d\omega$

e.g. $X(t) = \sin(\omega t + \theta)$, $\theta \sim U[-\pi, \pi]$

Find $S^{(Hz)}(f)$

$E(X(t)) = 0$ $\quad \cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$

$R(\tau) = E[X(t_1)X(t_2)]$ $\quad \sin\alpha\sin\beta = \frac{1}{2}[\cos(\alpha-\beta) - \cos(\alpha+\beta)]$

$= \frac{1}{2}[\cos(\omega(t_1 - t_2)) - \cos(\omega(\cancel{t_1 + t_2}) + 2\theta)]$

$= \frac{1}{2}\cos\omega\tau$

$S(\omega) = \frac{1}{2}\pi[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)]$ $\quad S^{(Hz)}$ 即 $S^{(rad)}$ 把 ω 换成 $2\pi f$

$S^{(Hz)}(f) = \frac{1}{4}[\delta(f-f_0) + \delta(f+f_0)]$ $\quad$ 而 $\delta(\omega)$ 要换成 $\frac{1}{2\pi}\delta(f)$

Linear System

$X(t) \longrightarrow \boxed{L_t} \longrightarrow y(t)$

$R_Y(t_1, t_2) = L_{t_1} L_{t_2} R_X(t_1, t_2)$

$X(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t)$

$Y(t) = h(t) * X(t)$

$R_Y(\tau) = h(\tau) * \bar{h}(-\tau) * R_X(\tau)$

$S_Y(\omega) = |H(\omega)|^2 S_X(\omega)$

$y(t) = ax(t)$, then $S_Y(\omega) = a^2 S_X(\omega)$

$R(\tau)$	$S(\omega)$
1	$2\pi\delta(\omega)$
$e^{j\omega_0\tau}$	$2\pi\delta(\omega-\omega_0)$
$\cos\omega_0\tau$	$\pi[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)]$
$e^{-\lambda\tau}u(\tau)$	$\frac{1}{j\omega+\lambda}$
$e^{-\lambda\|\tau\|}$	$\frac{2\lambda}{\omega^2+\lambda^2}$

$X(t), Y(t) \longrightarrow \otimes \longrightarrow Z(t)$ $\quad X(t)\cdot Y(t) \longrightarrow R_X(\tau)\cdot R_Y(\tau)$

$\longrightarrow S_X(\omega) * S_Y(\omega)$

e.g. $X(t)$ is WGN with $\frac{1}{2}N_0$, $h(t) = e^{-|t|}u(t)$

求 $R(\tau)$, $E(|Y(t)|^2)$

WGN, $S_X(\omega) = \frac{1}{2}N_0$

$H(\omega) = \frac{1}{1+j\omega}$, $|H(\omega)|^2 = \frac{1}{|1+j\omega|^2} = \frac{1}{1+\omega^2}$

$S_Y(\omega) = \frac{N_0}{2}\frac{1}{1+\omega^2}$

$R_Y(\tau) = \frac{N_0}{4}e^{-|\tau|}$

$E(|Y(t)|^2) = R(0) = \frac{N_0}{4}$